PULKIT TYAGI
Senior Gen-AI Engineer

ROADMAP TO BECOME AN AI ENGINEER IN 2026

This Is the Agentic AI Roadmap you need

🐍 Python & LLMs

🔗 Frameworks

🔍 RAG Systems

🤖 Agents

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STAGE 1: PYTHON FUNDAMENTALS

Master Python basics before diving into AI. Clean, readable code is the foundation.

1. **Core Syntax:** Variables, data types, functions, loops, conditionals.
2. **OOP Concepts:** Classes, objects, inheritance, encapsulation.
3. **API Handling:** HTTP requests, JSON parsing, async/await.
4. **Project Structure:** Modules, packages, virtual envs, requirements.txt.

PRO TIP

Focus on writing clean, readable code. Good habits save you later.

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STAGE 2: LLM FUNDAMENTALS

Understand high-level concepts. You don't need to train models, but you must know how to use them.

1. **Mechanics:** Tokens, next-token prediction, temperature, top-p.
2. **Context Windows:** Token limits (4K-128K), context management.
3. **Prompting:** Zero-shot, few-shot, chain-of-thought, system prompts.
4. **Structured Outputs:** JSON mode, tool calling, schema validation.

PRO TIP

Experiment with different models (GPT-5, Claude, Gemini). Each has strengths.

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STAGE 3: LEARN A FRAMEWORK

Choose ONE and master it. Depth beats breadth.

- (A) **LangChain + LangGraph:** Most popular, industry standard. (Recommended)
- (B) **CrewAI:** Focus on multi-agent orchestrations.
- (C) **Autogen:** Microsoft supported, powerful but complex.

PRO TIP

LangChain + LangGraph is the safest bet for maximum job opportunities.

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STAGE 4: ADVANCED CONCEPTS

Go deeper. Learn patterns that separate beginners from pros.

1. **State Management:** Track conversation state and context.
2. **Complex Workflows:** Conditional branching, loops, parallel paths.
3. **LCEL (LangChain):** Declarative composition for chains.
4. **Graph Architectures:** Nodes, edges, conditional routing.

PRO TIP

Build 2-3 projects using advanced features. Theory needs practice.

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STAGE 5: MEMORY MANAGEMENT

LLMs are stateless. Add memory to remember conversations.

1. **Short-term Memory:** Session history (Checkpointers).
2. **Long-term Memory:** User prefs across sessions (Databases).
3. **Thread IDs:** Unique identifiers for user conversations.
4. **Patterns:** Buffer, summary, entity extraction.

PRO TIP

Use LangGraph for memory—it's the modern approach.

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STAGE 6: TOOL INTEGRATION

Extend capabilities. Connect AI to APIs, search, and databases.

1. **@tool Decorator:** Convert functions to tools.
2. **Tool Schemas:** Define inputs/outputs for the LLM.
3. **External APIs:** Weather, search, custom business logic.
4. **Tool Selection:** How LLMs decide which tool to use.

PRO TIP

Write clear docstrings—this is how the LLM "reads" your tools.

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STAGE 7: RAG SYSTEMS

Retrieval Augmented Generation. Connect LLMs to custom data.

1. **Vector Stores:** Chroma, FAISS, Pinecone.
2. **Embeddings:** Converting text to semantic vectors.
3. **Retrieval:** Similarity search, MMR, hybrid search.
4. **Multi-Source:** PDFs, CSVs, Web scraping.

PRO TIP

RAG is the #1 enterprise use case. Master it well.

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STAGE 8: AGENTS & MULTI-AGENTS

Autonomous thinking. Agents reason, use tools, and act.

1. **ReAct Pattern:** Reason → Act → Observe → Repeat.
2. **Single Agents:** Focused tasks (coding, research).
3. **Multi-Agent:** Supervisor, worker, critic patterns.
4. **Use Cases:** When to use Agents vs Chains?

💡 **PRO TIP**

Start with single agents. Multi-agent systems are harder to debug.

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STAGE 9: REAL-WORLD PROJECTS

Apply your knowledge. Build end-to-end applications.

1. **Backend:** FastAPI to serve your Agents.
2. **Frontend:** Streamlit (demo) or React/Next.js (prod).
3. **Deployment:** Docker, AWS, or GCP.
4. **Databases:** Postgres (data) + Redis (cache).

PRO TIP

Deployed projects impress more than tutorials.

TAP THAT FOLLOW BUTTON AND STAY IN THE LOOP!

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